

Computational Mathematics for Learning and Data Analysis

Computational Foundations, Numerical Linear Algebra, and
Optimization

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Part I

**Numerical Linear Algebra and
Data Analysis**

Chapter 1

Foundations of Numerical Linear Algebra and SVD Decomposition

1.1 The Role of Linear Algebra in Machine Learning and Data Analysis

Modern machine learning and large-scale data analysis rely extensively on concepts and methods derived from numerical linear algebra and continuous optimization [1, 2]. Real-world data (images, texts, time series, graphs, or sensor measurements) are systematically encoded as numerical vectors and matrices:

- **Design Matrix:** Let $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be a set of n statistical samples, each described by d numerical features. The entire collection is organized as a matrix $X \in \mathbb{R}^{n \times d}$:

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}, \quad (1.1)$$

where rows represent observations and columns represent measured variables.

- **Gram Matrix and Covariance Matrix:** Linear relationships between pairs of observations or features are quantified through matrix products: the Gram matrix $G = XX^T \in \mathbb{R}^{n \times n}$ quantifies similarities

between data samples, whereas the uncentered sample covariance matrix $\Sigma_X = \frac{1}{n}X^T X \in \mathbb{R}^{d \times d}$ measures variance and pairwise covariance among features.

Unlike symbolic or purely theoretical linear algebra, *computational* or *numerical* linear algebra must address two fundamental constraints: asymptotic computational complexity for large-scale operations and error propagation stemming from finite-precision machine arithmetic (IEEE 754 standard) [3].

1.2 Vector and Matrix Norms

Quantifying the magnitude of vectors and matrices, as well as measuring discrepancies between predictions and ground-truth targets, requires the axiomatic definition of a **norm**.

Definition 1.2.1 (Vector Norm). A function $\|\cdot\| : \mathbb{R}^d \rightarrow \mathbb{R}$ is called a **norm** on $\mathbb{R}^d$ if it satisfies the following three properties for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $\alpha \in \mathbb{R}$:

1. **Non-negativity and definiteness:** $\|\mathbf{x}\| \geq 0$, with $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$;
2. **Absolute homogeneity:** $\|\alpha\mathbf{x}\| = |\alpha| \|\mathbf{x}\|$;
3. **Triangle inequality:** $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

1.2.1 The Family of ℓ_p -Norms

The vector norms most widely encountered in data science belong to the parameterized family of ℓ_p -norms (with $p \geq 1$):

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}. \quad (1.2)$$

Three fundamental special cases are:

1. **Manhattan Norm (ℓ_1):** $\|\mathbf{x}\|_1 = \sum_{i=1}^d |x_i|$. Widely used to promote sparsity in linear models (Lasso regularization).
2. **Euclidean Norm (ℓ_2):** $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^d x_i^2} = \sqrt{\mathbf{x}^T \mathbf{x}}$. Represents standard geometric Euclidean distance and is invariant under orthogonal rotations.

3. **Chebyshev Norm** (ℓ_∞): $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq d} |x_i|$. Captures peak deviation across any individual coordinate.

Property 1.2.1 (Topological Equivalence of Norms). *In a finite-dimensional real vector space $\mathbb{R}^d$, all norms are topologically equivalent: for any pair of norms $\|\cdot\|_a$ and $\|\cdot\|_b$, there exist positive constants $c_1, c_2 > 0$ such that:*

$$c_1 \|\mathbf{x}\|_b \leq \|\mathbf{x}\|_a \leq c_2 \|\mathbf{x}\|_b \quad \forall \mathbf{x} \in \mathbb{R}^d. \quad (1.3)$$

1.2.2 Matrix Norms and Induced Norms

To quantify the magnitude of a linear operator $A \in \mathbb{R}^{m \times n}$, matrix norms are employed. Two main classes are distinguished:

Frobenius Norm. Directly extends the Euclidean vector norm by treating the matrix as a flattened vector in $\mathbb{R}^{m \cdot n}$:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2} = \sqrt{\text{Tr}(A^T A)}. \quad (1.4)$$

Induced (or Subordinate) Operator Norms. Measure the maximum amplification that matrix A can apply to a non-zero vector with respect to a given vector norm:

$$\|A\|_p = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|A\mathbf{x}\|_p}{\|\mathbf{x}\|_p} = \max_{\|\mathbf{x}\|_p=1} \|A\mathbf{x}\|_p. \quad (1.5)$$

Prominent closed-form computational expressions include:

- $\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$ (maximum absolute column sum);
- $\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$ (maximum absolute row sum);
- $\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^T A)}$ (**spectral norm**, equal to the largest singular value of A).

Every induced matrix norm is **sub-multiplicative**: $\|AB\| \leq \|A\| \|B\|$.

1.3 Conditioning and Numerical Stability

A computational problem $y = f(x)$ is called **well-conditioned** if small perturbations in the input data x lead to proportionally small variations in the solution y . Conversely, a problem is **ill-conditioned** if slight disturbances in the inputs produce macroscopic fluctuations in the output [2].

1.3.1 Condition Number of a Linear System

Consider the square linear system $A\mathbf{x} = \mathbf{b}$, where $A \in \mathbb{R}^{n \times n}$ is non-singular and $\mathbf{b} \neq \mathbf{0}$. Suppose the right-hand side vector is corrupted by additive noise $\delta\mathbf{b}$, inducing a perturbation $\delta\mathbf{x}$ in the solution:

$$A(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b} \implies A\delta\mathbf{x} = \delta\mathbf{b} \implies \delta\mathbf{x} = A^{-1}\delta\mathbf{b}. \quad (1.6)$$

By the sub-multiplicativity of matrix norms:

$$\|\delta\mathbf{x}\| \leq \|A^{-1}\| \|\delta\mathbf{b}\| \quad \text{and} \quad \|\mathbf{b}\| \leq \|A\| \|\mathbf{x}\| \implies \frac{1}{\|\mathbf{x}\|} \leq \frac{\|A\|}{\|\mathbf{b}\|}. \quad (1.7)$$

Multiplying both inequalities yields the foundational relative error bound:

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq (\|A\| \|A^{-1}\|) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}. \quad (1.8)$$

Definition 1.3.1 (Condition Number). The **condition number** of an invertible square matrix $A \in \mathbb{R}^{n \times n}$ associated with norm $\|\cdot\|$ is:

$$\kappa(A) = \|A\| \|A^{-1}\|. \quad (1.9)$$

In the case of the spectral norm $\|\cdot\|_2$, letting $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$ denote the singular values of A :

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} = \frac{\sigma_1}{\sigma_n} \geq 1. \quad (1.10)$$

Computational Significance of $\kappa(A)$

In standard IEEE 754 double-precision arithmetic (machine precision $\epsilon_{\text{mach}} \approx 10^{-16}$), solving a linear system with $\kappa(A) \approx 10^k$ typically causes the loss of approximately k decimal digits of precision. If $\kappa(A) \geq 10^{16}$, the matrix is regarded as numerically singular.

1.4 Spectral Decomposition (Eigendecomposition)

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. A scalar $\lambda \in \mathbb{C}$ is an **eigenvalue** of A with associated non-zero **eigenvector** $\mathbf{v} \neq \mathbf{0}$ if it satisfies:

$$A\mathbf{v} = \lambda\mathbf{v}. \quad (1.11)$$

Theorem 1.4.1 (Spectral Theorem for Real Symmetric Matrices). *Let $A \in \mathbb{R}^{n \times n}$ be a real symmetric matrix ($A = A^T$). Then:*

1. *All eigenvalues of A are real: $\lambda_1, \dots, \lambda_n \in \mathbb{R}$.*
2. *Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.*
3. *There exists an orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ ($Q^T Q = Q Q^T = I_n$) whose columns are normalized eigenvectors of A , such that:*

$$A = Q \Lambda Q^T = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^T, \quad (1.12)$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

1.4.1 Symmetric Positive Semi-Definite (SPSD) Matrices and Quadratic Forms

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is defined as:

- **Positive semi-definite** ($A \succeq 0$) if $\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n \iff \lambda_i(A) \geq 0$ for all $i = 1, \dots, n$.
- **Positive definite** ($A \succ 0$) if $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0} \iff \lambda_i(A) > 0$ for all $i = 1, \dots, n$.

Example 1.4.1 (Sample Covariance Matrices are Always Positive Semi-Definite). Let $X \in \mathbb{R}^{n \times d}$ be a column-centered data matrix. The sample covariance matrix $C = \frac{1}{n} X^T X$ is symmetric and for any arbitrary vector $\mathbf{w} \in \mathbb{R}^d$:

$$\mathbf{w}^T C \mathbf{w} = \mathbf{w}^T \left(\frac{1}{n} X^T X \right) \mathbf{w} = \frac{1}{n} (X \mathbf{w})^T (X \mathbf{w}) = \frac{1}{n} \|X \mathbf{w}\|_2^2 \geq 0. \quad (1.13)$$

Therefore, all eigenvalues of C are real and non-negative, proving that $C \succeq 0$.

1.5 Singular Value Decomposition (SVD)

Spectral decomposition applies exclusively to symmetric square matrices. The **Singular Value Decomposition (SVD)** provides the fundamental generalization to arbitrary rectangular matrices $A \in \mathbb{R}^{m \times n}$ [2, 1].

Theorem 1.5.1 (Existence of the SVD). *Given any real matrix $A \in \mathbb{R}^{m \times n}$ of rank $r \leq \min(m, n)$, there exist two orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ and a pseudo-diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$ such that:*

$$A = U \Sigma V^T, \quad (1.14)$$

where the main diagonal of Σ contains the singular values arranged in non-increasing order:

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > \sigma_{r+1} = \dots = \sigma_{\min(m,n)} = 0. \quad (1.15)$$

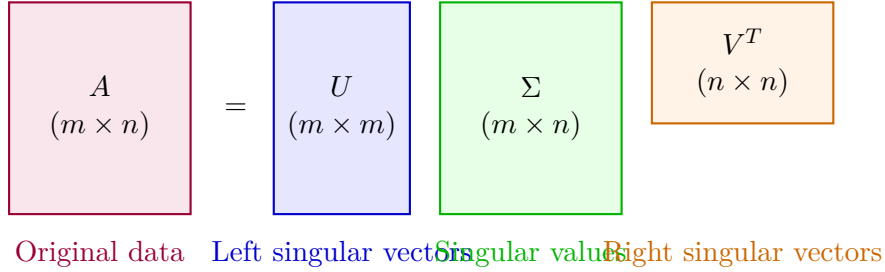


Figure 1.1: Dimensional block representation of the Singular Value Decomposition (Full SVD).

1.5.1 Outer Product Expansion and Thin SVD

The matrix A can be decomposed into an outer-product sum of r rank-one matrices:

$$A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T, \quad (1.16)$$

where:

- $\mathbf{u}_i \in \mathbb{R}^m$ is the i -th column of U (**left singular vector**), an eigenvector of AA^T with eigenvalue $\lambda_i = \sigma_i^2$;
- $\mathbf{v}_i \in \mathbb{R}^n$ is the i -th column of V (**right singular vector**), an eigenvector of $A^T A$ with eigenvalue $\lambda_i = \sigma_i^2$;
- $\sigma_i = \sqrt{\lambda_i(A^T A)} = \sqrt{\lambda_i(AA^T)}$ is the i -th singular value.

In computational settings where $m \gg n$, the **thin SVD** (or economy SVD) is used instead: it retains only the first n columns of U and the square diagonal block $\Sigma_n \in \mathbb{R}^{n \times n}$, reducing memory footprint from $\mathcal{O}(m^2)$ down to $\mathcal{O}(mn)$.

1.6 Properties and Applications of the SVD

1.6.1 Best Low-Rank Approximation: Eckart-Young-Mirsky Theorem

In exploratory data analysis, signal denoising, and lossy compression, one frequently seeks a low-rank matrix $\hat{A}_k$ with rank $k < r$ that best approximates the original matrix A .

Theorem 1.6.1 (Eckart-Young-Mirsky Theorem). *Let $A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ be the SVD of $A \in \mathbb{R}^{m \times n}$. For any integer $1 \leq k < r$, the truncated SVD matrix:*

$$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T = U_k \Sigma_k V_k^T \quad (1.17)$$

is the unique optimal solution to the low-rank approximation problem in both the spectral norm and the Frobenius norm:

$$A_k = \arg \min_{B \in \mathbb{R}^{m \times n}, \text{rank}(B) \leq k} \|A - B\|_2 = \arg \min_{B \in \mathbb{R}^{m \times n}, \text{rank}(B) \leq k} \|A - B\|_F. \quad (1.18)$$

Moreover, the optimal approximation errors are given explicitly by:

$$\|A - A_k\|_2 = \sigma_{k+1}, \quad (1.19)$$

$$\|A - A_k\|_F = \sqrt{\sum_{j=k+1}^r \sigma_j^2}. \quad (1.20)$$

1.6.2 The Moore-Penrose Pseudoinverse and Linear Least Squares

When a rectangular matrix $A \in \mathbb{R}^{m \times n}$ lacks a standard inverse ($m \neq n$ or rank-deficient), the SVD yields the canonical formulation of the **Moore-Penrose pseudoinverse** $A^\dagger \in \mathbb{R}^{n \times m}$:

$$A^\dagger = V \Sigma^\dagger U^T = \sum_{i=1}^r \frac{1}{\sigma_i} \mathbf{v}_i \mathbf{u}_i^T, \quad (1.21)$$

where $\Sigma^\dagger$ is obtained by transposing Σ and inverting only the strictly positive diagonal entries $\sigma_i > 0$.

Property 1.6.1 (Least Squares Property). *For the linear least squares problem $\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|_2$, the vector*

$$\mathbf{x}^* = A^\dagger \mathbf{b} \quad (1.22)$$

is the unique minimum-norm solution $\|\mathbf{x}^\|_2$ among all vectors minimizing the residual sum of squares.*

1.6.3 Connection Between SVD and Principal Component Analysis (PCA)

Let $X \in \mathbb{R}^{n \times d}$ denote a column-centered data matrix (empirical sample mean $\boldsymbol{\mu} = \mathbf{0}$). The empirical covariance matrix is:

$$C = \frac{1}{n-1} X^T X. \quad (1.23)$$

Expressing the SVD of X as $X = U \Sigma V^T$, we obtain:

$$C = \frac{1}{n-1} (V \Sigma U^T) (U \Sigma V^T) = \frac{1}{n-1} V \Sigma^2 V^T = V \left(\frac{\Sigma^2}{n-1} \right) V^T. \quad (1.24)$$

From this identity, three crucial insights follow directly:

1. The right singular vectors $\mathbf{v}_i$ of X (columns of V) are precisely the **eigenvectors of the sample covariance matrix** (PCA principal directions).
2. The variance explained by the i -th principal component is directly proportional to the squared singular value: $\lambda_i = \frac{\sigma_i^2}{n-1}$.
3. Low-dimensional data projections (*principal component scores*) onto the top k components can be obtained directly without ever instantiating $C \in \mathbb{R}^{d \times d}$:

$$Z_k = X V_k = (U_k \Sigma_k V_k^T) V_k = U_k \Sigma_k. \quad (1.25)$$

Listing 1.1: Compact Truncated SVD and PCA computation using NumPy.

```
1 import numpy as np
2
3 def compute_svd_pca(X: np.ndarray, k: int):
4     """
5     Computes a k-dimensional PCA projection via the
6     economy SVD.
7     X: data matrix of shape (n_samples, n_features)
8     k: number of principal components to retain
9     """
10    # 1. Center the data (zero-mean along columns)
11    X_centered = X - np.mean(X, axis=0)
12    n = X.shape[0]
13    # 2. Compute economy SVD
```

```

14     U, s, Vt = np.linalg.svd(X_centered, full_matrices=
15         False)
16
17     # 3. Truncate to top-k components
18     Uk = U[:, :k]
19     sk = s[:k]
20     Vk = Vt[:k, :].T
21
22     # 4. Compute projected coordinates (scores) and
23     explained variance
24     scores = Uk * sk # equivalent to X_centered @ Vk
25     explained_variance = (sk ** 2) / (n - 1)
26     explained_variance_ratio = explained_variance / np.
27         sum((s ** 2) / (n - 1))
28
29     return scores, Vk, explained_variance_ratio

```

1.7 Summary of Core Mathematical Concepts

Mathematical Tool	Operational Definition	Impact in Data Analysis / ML
ℓ_2 norm and $\ A\ _F$	Square root of sum of squared entries	Least squares optimization, MSE risk metric.
ℓ_1 norm	Sum of absolute values	Sparsity induction, robust feature selection (Lasso).
Condition number κ	$\sigma_{\max}/\sigma_{\min}$	Numerical stability, noise amplification in data.
Eigendecomposition	$A = Q\Lambda Q^T$ (only for $A = A^T$)	Spectral analysis, quadratic forms, positive kernels.
Full SVD	$A = U\Sigma V^T$ (arbitrary rectangular)	Universal geometric decomposition of data matrices.
Eckart-Young-Mirsky	$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$	Optimal low-rank compression, noise filtering.
SVD for PCA	$Z_k = U_k \Sigma_k$	Dimension reduction without forming $X^T X$.

Table 1.1: Summary of numerical linear algebra and SVD concepts covered in this chapter.

Part II

Continuous Optimization and Machine Learning

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